



INTRODUCTORY COMMUTATIVE ALGEBRA



Lecture Notes & Problem Discussions

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ABSTRACT

These are lecture notes for a first course in Commutative Algebra, covering rings, ideals, and modules, together with the accompanying problem-set discussions worked out alongside the lectures. The standard references for this material are [\[AM69\]](#), [\[Bly90\]](#), and [\[DF04\]](#).

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CHAPTER 1

LECTURE NOTES

1.1 Lecture 1

Convention. Rings are commutative with 1.

Definition 1.1. Let R, S be rings. A *morphism* φ from R to S is a map $\varphi : R \rightarrow S$ such that

- (1) $\varphi(r + s) = \varphi(r) + \varphi(s)$,
- (2) $\varphi(rs) = \varphi(r)\varphi(s)$,
- (3) $\varphi(1) = 1$.

Convention. A subring R must contain 1.

Question. What is $R[X]$?

X is not an element of R . Consider

$$S = \{f : \mathbb{N} \rightarrow R \mid f(r) = 0 \text{ for all but finitely many } r\},$$

with the usual addition and multiplication in sequences (Cauchy product). Then S is a ring with identity

$$1_S = (1, 0, 0, \dots, 0).$$

Define $\varphi : R \rightarrow S$ by $\varphi(a) = (a, 0, \dots, 0)$. Then φ is a ring homomorphism, for sure.

Define $X = (0, 1, 0, \dots)$ and observe $X^2 = (0, 0, 1, 0, \dots)$. A polynomial P over R can then be identified as an element of S , say

$$P = (P_0, P_1, \dots, P_n, 0, \dots) = \varphi(P_0) \cdot 1_S + \varphi(P_1) \cdot X + \dots + \varphi(P_n) \cdot X^n.$$

Definition 1.2. Let R be a ring. A non-zero, non-unit element $p \in R$ is *prime* if

$$p \mid ab \implies p \mid a \text{ or } p \mid b$$

(include the elementwise definition: for $a, b \in R$, $a \mid b$ if $\exists r \in R$ such that $b = r \cdot a$).

Definition 1.3. Let R be a ring. A proper ideal P of R is called a *prime ideal* if

$$ab \in P \implies a \in P \text{ or } b \in P.$$

Definition 1.4. Let R be a ring. $I \subseteq R$ is called an *ideal* if there exists a ring S and a morphism $\varphi : R \rightarrow S$ such that $I = \ker(\varphi)$.

Remark 1.5. This immediately gives the elementwise definition of ideals.

$$\mathcal{N}(R) = \{\text{all nilpotent elements of } R\} \rightsquigarrow \text{the nil-radical of } R.$$

Theorem 1.6.

$$\mathcal{N}(R) = \bigcap_{P \in \text{Spec } R} P, \quad \text{where } \text{Spec } R = \text{all prime ideals of } R.$$

Proof. Let $x \in \mathcal{N}(R)$. Then for any $P \in \text{Spec } R$, there exists $n_P \in \mathbb{N}$ such that

$$x^{n_P} = 0 \implies 0 \in P \implies x^{n_P} \in P \implies x \in P.$$

Now let $x \notin \mathcal{N}(R)$, and define

$$S = \{I \subseteq R \text{ ideal} \mid x^n \notin I \text{ for all } n\}.$$

$S \neq \emptyset$, as $\langle 0 \rangle \in S$, and S is a poset (ordered by inclusion). By Zorn's Lemma, let M be a maximal element of S .

Claim. M is a prime ideal.

Proof of claim. Let $a \notin M$, $b \notin M$. Now,

$$M + \langle a \rangle, M + \langle b \rangle \notin S \implies \exists m, n \in \mathbb{N} \text{ such that}$$

$$x^m \in M + \langle a \rangle \text{ and } x^n \in M + \langle b \rangle \implies x^{m+n} \in M + \langle ab \rangle.$$

If $ab \in M$, then $x^{m+n} \in M$, a contradiction. Hence $ab \notin M$, so M is prime. □

Since $x \notin M$ and $M \in \text{Spec } R$, we conclude $x \notin \bigcap_{P \in \text{Spec } R} P$, completing the proof. □

Question. What are the units of $R[X]$?

If R is an integral domain, then they are precisely the units of R .

Proposition 1.7. $f(X) = a_n X^n + \cdots + a_1 X + a_0$ is a unit if a_0 is a unit and the a_i 's are nilpotent.

Proof. Let $P \in \text{Spec } R$. We have a natural morphism $\varphi : R[X] \rightarrow (R/P)[X]$ given by

$$\varphi(b_m X^m + \cdots + b_1 X + b_0) = \overline{b_m} X^m + \cdots + \overline{b_1} X + \overline{b_0}, \quad (\overline{b_i} = b_i \bmod P).$$

Then $\varphi(f)$ is a unit in $(R/P)[X]$, and since R/P is an integral domain, this forces

$$\overline{a_i} \equiv \overline{0} \pmod{P} \text{ for all } i \in [n], \quad \text{and} \quad \overline{a_0} \text{ is a unit of } R/P.$$

That is, $a_0 \notin P$ and $a_1, \dots, a_n \in P$. As $P \in \text{Spec } R$ was arbitrary,

$$a_0 \notin P \text{ and } a_i \in P \quad \forall i \in [n], \quad \text{for all } P \in \text{Spec } R.$$

The other direction is trivial. □

1.2 Lecture 2

Definition 1.8. Let R be a ring and let $(M, +)$ be an abelian group. A *scalar multiplication*

$$R \times M \longrightarrow M, \quad (r, m) \longmapsto rm,$$

makes M into an R -module if the following hold for all $r, s \in R$ and $m, m_1, m_2 \in M$:

- (i) $(r + s)(m) = rm + sm$,
- (ii) $r(m_1 + m_2) = rm_1 + rm_2$,
- (iii) $r(sm) = rsm = s(rm)$,
- (iv) $1 \cdot m = m$.

Definition 1.9. If M, N are R -modules, a map $\varphi : M \rightarrow N$ is called an R -module morphism if

- (i) $\varphi(m_1 + m_2) = \varphi(m_1) + \varphi(m_2)$,
- (ii) $\varphi(rm) = r\varphi(m)$.

Definition 1.10. Let R be a ring and S a nonempty set. A *free R -module on S* is a pair (F, f) , where F is an R -module and $f : S \rightarrow F$ is a set-theoretic map, such that: for any R -module M and any set-theoretic map $g : S \rightarrow M$, there exists a unique R -module morphism $h : F \rightarrow M$

such that

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$

commutes, i.e. $h \circ f = g$.

Proposition 1.11. Let (F, f) be a free R -module on S . Then:

- (1) f is injective;
- (2) $\text{Im}(f)$ generates F as an R -module.

Proof of (1). Let $x, y \in S$ with $x \neq y$. **Goal:** $f(x) \neq f(y)$.

Take any R -module M with at least two elements, and define $g : S \rightarrow M$ such that $g(x) \neq g(y)$. By definition, there exists a (unique) morphism $h : F \rightarrow M$ such that $h \circ f = g$. Then

$$(h \circ f)(x) = g(x) \neq g(y) = (h \circ f)(y) \implies f(x) \neq f(y). \quad \square$$

Definition 1.12. Let M be an R -module and $\emptyset \neq A \subseteq M$. The *submodule of M generated by A* is, equivalently:

- (1) the smallest submodule of M containing A ;
- (2) the intersection of all submodules of M containing A ;
- (3) the set of all finite R -linear combinations of elements of A .

Proof of (2). Let A be the submodule of F generated by $\text{Im}(f)$. Since $f(s) \in \text{Im}(f) \subseteq A$ for every $s \in S$, f factors as $f = i_A \circ f^*$ for a set-theoretic map $f^* : S \rightarrow A$ and the inclusion $i_A : A \hookrightarrow F$:

$$\begin{array}{ccc} S & \xrightarrow{f^*} & A \\ & \searrow f & \downarrow i_A \\ & & F \end{array}$$

As (F, f) is free on S and A is an R -module, there is a unique morphism $h : F \rightarrow A$ such that $h \circ f = f^*$:

$$\begin{array}{ccc} S & \xrightarrow{f^*} & A \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$

It is enough to show i_A is onto. We compute

$$(i_A \circ h) \circ f = i_A \circ (h \circ f) = i_A \circ f^* = f.$$

So $i_A \circ h$ and id_F both satisfy $(-) \circ f = f$; by uniqueness in the definition of freeness (applied with $M = F, g = f$), $i_A \circ h = \text{id}_F$. Hence i_A is onto, i.e. $A = F$, so $\text{Im}(f)$ generates F . $\square$

Theorem 1.13: Uniqueness of free modules. Let (F, f) be a free R -module on a set S . Then (F', f') is also a free R -module on S if and only if there exists a unique R -module isomorphism $j : F \rightarrow F'$ such that $j \circ f = f'$.

Proof. Suppose (F', f') is also free on S . Since (F, f) is free (applied to $M = F', g = f'$), there is a morphism $h : F \rightarrow F'$ with $h \circ f = f'$:

$$\begin{array}{ccc} S & \xrightarrow{f'} & F' \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$

Likewise, since (F', f') is free (applied to $M = F, g = f$), there is a morphism $h' : F' \rightarrow F$ with $h' \circ f' = f$:

$$\begin{array}{ccc} S & \xrightarrow{f} & F \\ f' \downarrow & \nearrow h' & \\ F' & & \end{array}$$

Then $(h' \circ h) \circ f = h' \circ (h \circ f) = h' \circ f' = f$. So $h' \circ h$ and id_F both satisfy $(-) \circ f = f$; by uniqueness (with $M = F, g = f$), $h' \circ h = \text{id}_F$. Similarly $h \circ h' = \text{id}_{F'}$. So $j := h$ is an isomorphism with $j \circ f = f'$.

Converse. Suppose there is an isomorphism $j : F \rightarrow F'$ with $j \circ f = f'$. To show (F', f') is free on S , take any R -module M and set-theoretic map $g : S \rightarrow M$; we must produce a unique $h' : F' \rightarrow M$ with $h' \circ f' = g$:

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f' \downarrow & \nearrow h' & \\ F' & & \end{array}$$

Since (F, f) is free, there is a unique $h : F \rightarrow M$ with $h \circ f = g$. From $j \circ f = f'$ we get $j^{-1} \circ f' = f$, so

$$(h \circ j^{-1}) \circ f' = h \circ (j^{-1} \circ f') = h \circ f = g,$$

i.e. $h' := h \circ j^{-1}$ works:

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f' \downarrow & \searrow f & \uparrow h \\ F' & \xrightarrow{j^{-1}} & F \end{array}$$

Uniqueness. Suppose $t : F' \rightarrow M$ also satisfies $t \circ f' = g$. Then $t \circ j : F \rightarrow M$ satisfies $(t \circ$

$j \circ f = t \circ (j \circ f) = t \circ f' = g$, so by uniqueness of h (from freeness of (F, f)), $t \circ j = h$, hence $t = h \circ j^{-1} = h'$. $\square$

Theorem 1.14: Existence of free R -modules. For any ring R and nonempty set S , a free R -module on S exists.

Proof. Define

$$F = \{\theta : S \rightarrow R \mid \theta(s) = 0 \text{ for almost all } s \in S\}.$$

For $\theta_1, \theta_2 \in F$ and $\lambda \in R$, define

$$(\theta_1 + \theta_2)(s) = \theta_1(s) + \theta_2(s), \quad (\lambda\theta)(s) = \lambda\theta(s).$$

One checks that F is an R -module. Define $f : S \rightarrow F$ by $f(s) = \delta_s$, where $\delta_s(t) = 1$ if $t = s$ and 0 otherwise.

Existence of h . Let M be an R -module and $g : S \rightarrow M$ a set-theoretic map. Define $h : F \rightarrow M$ by

$$h(\theta) = \sum_{s \in S} \theta(s) g(s),$$

a finite sum for every $\theta \in F$. For any $t \in S$,

$$(h \circ f)(t) = h(f(t)) = h(\delta_t) = \delta_t(t) g(t) = g(t),$$

so $h \circ f = g$:

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$

Uniqueness of h . For $\theta \in F$ and $t \in S$,

$$\theta(t) = \theta(t) \cdot 1_R = \sum_{s \in S} \theta(s) \delta_s(t) = \left(\sum_{s \in S} \theta(s) \delta_s \right) (t) \quad \forall t \in S,$$

so $\theta = \sum_{s \in S} \theta(s) \delta_s$. Now let $h' : F \rightarrow M$ be any R -module morphism with $h' \circ f = g$. Then for $\theta \in F$,

$$h'(\theta) = h' \left(\sum_{s \in S} \theta(s) \delta_s \right) = \sum_{s \in S} \theta(s) h'(\delta_s) = \sum_{s \in S} \theta(s) h'(f(s)) = \sum_{s \in S} \theta(s) g(s) = h(\theta).$$

Hence $h = h'$. $\square$

1.3 Lecture 3

Theorem 1.15. Let R be a ring such that every prime ideal is principal. Then every ideal is principal.

Proof. Let $S = \{\text{all non-principal ideals of } R\}$. For the sake of contradiction (FTSOC), suppose $S \neq \emptyset$. By Zorn's Lemma, let I be a maximal element of S .

Since I is not principal, I is not prime by hypothesis, so there exist $a, b \in R$ with $a \notin I, b \notin I$, but $ab \in I$.

Let $J = \langle a \rangle + I \supsetneq I$; since I is maximal among non-principal ideals, $J \notin S$, so J is principal, say $J = \langle c \rangle$.

Also let $K = \{r \in R \mid ra \in I\} \supsetneq I$ (since $b \in K \setminus I$); again $K \notin S$, so $K = \langle d \rangle$ for some d .

$I \subseteq \langle cd \rangle$. Let $x \in I$. Since $I \subseteq J = \langle c \rangle$, there is $\lambda \in R$ with $\lambda c = x$. Now $\lambda a \in I$, so $\lambda \in K = \langle d \rangle$, and hence $x \in \langle cd \rangle$. Thus $I \subseteq \langle cd \rangle$.

$\langle cd \rangle \subseteq I$. There exist $u \in R, v \in I$ such that $c = ua + v$ (since c generates $J = \langle a \rangle + I$). Also $da \in I$ (since d generates K). Then

$$cd = (ua + v)d = u(ad) + vd \in I,$$

since $ad = da \in I$ and $v \in I$. Hence $\langle cd \rangle \subseteq I$.

Together, $I = \langle cd \rangle$, contradicting $I \in S$. So $S = \emptyset$, i.e. every ideal of R is principal. $\square$

Theorem 1.16. Let R be a ring such that every prime ideal is finitely generated. Then every ideal of R is finitely generated.

Proof. Let $A = \{I \text{ ideal of } R \mid I \text{ is not finitely generated}\}$. By Zorn's Lemma, let M be a maximal element of A .

Claim: M is a maximal ideal. Suppose not, so there is an ideal N with $M \subsetneq N \subsetneq R$. Since N is maximal (hence prime), it is finitely generated by hypothesis: there exist $a_1, \dots, a_n \in R$ such that $N = \langle a_1, \dots, a_n \rangle$.

Since $M \subsetneq N$, some $a_i \notin M$. Then $M + \langle a_i \rangle \supsetneq M$, so by maximality of M in A , $M + \langle a_i \rangle \notin A$; hence there exist $b_1, \dots, b_k \in R$ such that

$$M + \langle a_i \rangle = \langle b_1, \dots, b_k \rangle.$$

$\square$

Lemma 1.17. Let I, J_1, J_2 be ideals of R such that $I \subseteq J_1 \cup J_2$. Then either $I \subseteq J_1$ or $I \subseteq J_2$.

Proof. FTSOC, suppose $I \setminus J_1 \neq \emptyset$ and $I \setminus J_2 \neq \emptyset$. Let $x \in I \setminus J_1$ and $y \in I \setminus J_2$; since $I \subseteq J_1 \cup J_2$, this forces $x \in J_2$ and $y \in J_1$. Also $x + y \in I \subseteq J_1 \cup J_2$, so $x + y \in J_1$ or $x + y \in J_2$; WLOG $x + y \in J_1$. Since $y \in J_1$, $x = (x + y) - y \in J_1$, contradicting $x \notin J_1$. $\square$

Theorem 1.18: Prime Avoidance. Let $I, J_1, \dots, J_n$ be ideals of R such that

- (1) $I \subseteq J_1 \cup \dots \cup J_n$,
- (2) $J_3, \dots, J_n \in \text{Spec } R$.

Then $I \subseteq J_i$ for some i .

Proof. We induct on n . The case $n = 2$ needs no primality hypothesis and is the lemma above; assume $n \geq 3$ and the result for any collection of $n - 1$ such ideals.

We prove the contrapositive: if $I \not\subseteq J_i$ for every i , then $I \not\subseteq \bigcup_{i=1}^n J_i$.

Fix i . Apply the induction hypothesis to I together with the $n - 1$ ideals $\{J_j\}_{j \neq i}$ (which still has all but at most two members prime): if $I \subseteq \bigcup_{j \neq i} J_j$, the induction hypothesis would give $I \subseteq J_j$ for some $j \neq i$, contradicting our assumption. So $I \not\subseteq \bigcup_{j \neq i} J_j$, and we may choose

$$z_i \in I \setminus \bigcup_{j \neq i} J_j.$$

Since $I \subseteq \bigcup_{i=1}^n J_i$ by hypothesis (1), and $z_i \notin J_j$ for every $j \neq i$, we must have $z_i \in J_i$, for every i .

Set

$$z := z_1 z_2 \cdots z_{n-1} + z_n \in I$$

(a sum of an element of I and a product of elements of I , hence itself in I). We show $z \notin J_i$ for every i , which gives $I \not\subseteq \bigcup_{i=1}^n J_i$, as desired.

- If $i \in \{1, \dots, n - 1\}$: since $z_i \in J_i$ is one of the factors of $z_1 \cdots z_{n-1}$, this product lies in J_i . If $z \in J_i$, then $z_n = z - z_1 \cdots z_{n-1} \in J_i$, contradicting $z_n \notin J_i$ (as $i \neq n$).
- If $i = n$: if $z \in J_n$, then since $z_n \in J_n$ as well, $z_1 \cdots z_{n-1} = z - z_n \in J_n$. As J_n is prime, some factor $z_k \in J_n$ for a $k \in \{1, \dots, n - 1\}$, contradicting $z_k \notin J_n$ (as $k \neq n$).

This completes the induction. □

Proposition 1.19. An element of $\mathbb{Q}(i)$ lies in $\mathbb{Z}[i]$ if and only if it is the root of a monic polynomial over $\mathbb{Z}$.

Proof. Let $a + bi \in \mathbb{Q}(i)$. Then $a + bi$ is a root of

$$P(X) = X^2 - 2aX + (a^2 + b^2) \in \mathbb{Q}[X].$$

Claim: $2a \in \mathbb{Z}$ and $a^2 + b^2 \in \mathbb{Z}$ together imply $a, b \in \mathbb{Z}$.

Since $2a \in \mathbb{Z}$, $(2a)^2 \in \mathbb{Z}$. Since $a^2 + b^2 \in \mathbb{Z}$, $4(a^2 + b^2) = (2a)^2 + (2b)^2 \in \mathbb{Z}$, so $(2b)^2 = 4(a^2 + b^2) - (2a)^2 \in \mathbb{Z}$. As $2b \in \mathbb{Q}$ has integer square, $2b \in \mathbb{Z}$.

Now $(2a)^2 + (2b)^2 = 4(a^2 + b^2) \equiv 0 \pmod{4}$. Since the square of any integer is $\equiv 0 \pmod{4}$ if it

is even, and $\equiv 1 \pmod{4}$ if it is odd, the only way for the sum to be $\equiv 0 \pmod{4}$ is

$$(2a)^2 \equiv (2b)^2 \equiv 0 \pmod{4},$$

forcing $2a, 2b$ to both be even, i.e. $a, b \in \mathbb{Z}$.

Since $a + bi \in \mathbb{Z}[i]$ exactly when $a, b \in \mathbb{Z}$, and $a + bi$ is a root of a monic polynomial over $\mathbb{Z}$ exactly when $2a, a^2 + b^2 \in \mathbb{Z}$ (the coefficients of $P(X)$), the claim gives the equivalence. $\square$

Definition 1.20. Let $R \hookrightarrow S$ be rings. An element $\alpha \in S$ is called *integral* over R if α is a root of a monic polynomial

$$X^n + a_{n-1}X^{n-1} + \cdots + a_0$$

over R (i.e. with $a_0, \dots, a_{n-1} \in R$).

1.4 Lecture 4

Definition 1.21. Let R be a subring of S , and let $U \subseteq S$ be a subset. Define

$$R[U] := \text{the smallest subring of } S \text{ containing } R \text{ and } U.$$

Remark 1.22. For $U, V \subseteq S$,

$$R[U][V] = R[U \cup V] = R[V][U].$$

In particular, if $V = \{u_1, \dots, u_n\}$, then

$$R[u_1, \dots, u_n] = R[u_1][u_2] \cdots [u_n].$$

Proposition 1.23.

$$R[u] = \{a_0 + a_1u + \cdots + a_nu^n \mid a_i \in R, n \in \mathbb{N}_{\geq 0}\}.$$

Proof. Let $T = \{a_0 + a_1u + \cdots + a_nu^n \mid a_i \in R, n \in \mathbb{N}_{\geq 0}\}$.

T is a **subring** of S . It contains 1 (take $n = 0, a_0 = 1$). It is closed under addition: given two such expressions, pad the shorter one with zero coefficients to a common degree and add coefficientwise, which stays in R since R is a ring. It is closed under multiplication: expanding a product of two such expressions and collecting powers of u , each resulting coefficient is a finite sum of products of elements of R , hence lies in R .

$R[u] \subseteq T$. Since $R \subseteq T$ (take $n = 0$) and $u \in T$ (take $n = 1, a_0 = 0, a_1 = 1$), and T is a subring of S containing R and u , while $R[u]$ is by definition the *smallest* such subring, we get $R[u] \subseteq T$.

$T \subseteq R[u]$. Since $R[u]$ is a subring containing R and u , it is closed under addition and multiplication. So for any $a_0, \dots, a_n \in R \subseteq R[u]$, each $a_iu^i \in R[u]$, and hence $a_0 + a_1u + \cdots + a_nu^n \in R[u]$.

Combining both containments, $R[u] = T$. □

Example 1.24. The map $\mathbb{Z}[X] \rightarrow \mathbb{Z}[i]$, $X \mapsto i$, is a surjective ring morphism with kernel $\langle X^2 + 1 \rangle$.

Proposition 1.25. Let R, S be rings and $\varphi : R \rightarrow S$ be a morphism. Let $\alpha \in S$ be an element. Then φ can be uniquely extended to a morphism $\tilde{\varphi} : R[X] \rightarrow S$ such that $\tilde{\varphi}|_R = \varphi$ and $\tilde{\varphi}(X) = \alpha$.

Proof sketch. Define $\tilde{\varphi} : R[X] \rightarrow S$ by

$$\tilde{\varphi}(a_0 + a_1X + \cdots + a_nX^n) = \varphi(a_0) + \varphi(a_1)\alpha + \cdots + \varphi(a_n)\alpha^n,$$

and $\tilde{\varphi}(A \cdot B) = \tilde{\varphi}(A) \cdot \tilde{\varphi}(B)$. □

Let R be a subring of S , and let $u \in S$. Then there is a unique morphism $\varphi : R[X] \rightarrow S$ such that $\varphi|_R = \text{id}_R$ and $\varphi(X) = u$ (apply the above with $\varphi = \text{id}_R$, $\alpha = u$). Then $\text{Im}(\varphi) = R[u]$; letting $I = \ker \varphi$,

$$R[X]/I \cong R[u].$$

Notation. $I \cap R = \langle 0 \rangle$.

Conversely, let I be an ideal of $R[X]$ such that $I \cap R = \langle 0 \rangle$. We have the natural projection map

$$R[X] \longrightarrow R[X]/I.$$

Since $I \cap R = 0$, $R \hookrightarrow R[X]/I$, so we can identify $a \in R$ with its image $a + I$.

Question. What is the image of $R[X]$ in $R[X]/I$? (It will be determined by $\text{Im}(R)$ and that of X .)

Under the projection, $X \mapsto X + I$. Writing $u = X + I$,

$$R[X]/I \cong R[u].$$

Definition 1.26. Let $R \hookrightarrow S$. S is said to be an *integral extension* of R if each $\alpha \in S$ is integral over R .

Notation. (Recall.) Let R be a commutative ring with 1, and let $A \in M_n(R)$. Then

$$A \cdot \text{adj}(A) = \text{adj}(A) \cdot A = \det(A) \cdot I_n.$$

Theorem 1.27. Let $R \hookrightarrow S$ and $b \in S$. Then

$$b \text{ is integral over } R \iff R[b] \text{ is a finitely generated } R\text{-module.}$$

Proof. ($\Rightarrow$) Assume b is integral over R ; then there is a monic polynomial $f(X) \in R[X]$ such that

$f(b) = 0$. Let $\deg f = n \geq 1$. Take any $g(b) \in R[b]$. By the division algorithm,

$$g(X) = q(X)f(X) + r(X), \quad \deg r < n,$$

so $g(b) = r(b)$. Hence $R[b]$ is generated (as an R -module) by $\{1, b, \dots, b^{n-1}\}$.

($\Leftarrow$) Conversely, let $R[b]$ be a finitely generated R -module, say $R[b] = \langle w_1, \dots, w_n \rangle$. Then for each $i \in [n]$,

$$bw_i = a_{i1}w_1 + \dots + a_{in}w_n, \quad a_{ij} \in R,$$

i.e. $(bI - A)\vec{w} = 0$, where $A = (a_{ij})$. Multiplying by $\text{adj}(bI - A)$,

$$\det(bI - A)w_i = 0 \quad \forall i \in [n].$$

Since $1 \in R[b] = \langle w_1, \dots, w_n \rangle$, we get $\det(bI - A) = 0$, which is a monic equation in b (of degree n). Hence b is integral over R . $\square$

Theorem 1.28. Let $R \hookrightarrow S$, and let $b_1, b_2, \dots, b_n \in S$. Then each b_i is integral over R if and only if $R[b_1, \dots, b_n]$ is a finitely generated R -module.

Proof. Induct on n . The case $n = 1$ is the previous theorem.

For $n > 1$: assume $b_1, \dots, b_n$ are integral over R . In particular b_n is integral over R , hence also integral over $R[b_1, \dots, b_{n-1}]$; by the previous theorem (applied over the base ring $R[b_1, \dots, b_{n-1}]$), $R[b_1, \dots, b_{n-1}][b_n]$ is a finitely generated $R[b_1, \dots, b_{n-1}]$ -module, which (combined with the induction hypothesis that $R[b_1, \dots, b_{n-1}]$ is a finitely generated R -module) makes $R[b_1, \dots, b_n]$ a finitely generated R -module.

The converse is the same as the $n = 1$ case. $\square$

Exercise 1.29. Let $R \hookrightarrow S$, and let $a, b \in S$ be integral over R . Then $a \pm b$ and ab are integral over R .

1.5 Lecture 5

Corollary 1.30. Let $R \hookrightarrow S$ and $b_1, \dots, b_n \in S$ be integral over R (equivalently, $R[b_1, \dots, b_n]$ is a finitely generated R -module). Then every $b \in R[b_1, \dots, b_n]$ is integral over R .

Proof. Since $b \in R[b_1, \dots, b_n]$,

$$R[b_1, \dots, b_n] = R[b_1, \dots, b_n, b],$$

which is a finitely generated R -module. By the previous theorem, b is integral over R . $\square$

Proposition 1.31. Let $A \subseteq B \subseteq C$ be rings such that B is integral over A and C is integral over B . Then C is integral over A .

Proof. Let $c \in C$. Then c is integral over B :

$$c^n + b_{n-1}c^{n-1} + \cdots + b_0 = 0, \quad b_i \in B.$$

Let $R := A[b_0, \dots, b_{n-1}]$. Then c is integral over R , so $R[c]$ is a finitely generated R -module.

Since $b_0, \dots, b_{n-1}$ are integral over A , R is a finitely generated A -module. Hence $R[c]$ is a finitely generated A -module, and since $c \in R[c]$, c is integral over A . $\square$

Definition 1.32. Define $\bar{R}^S := \{r \in S \mid r \text{ is integral over } R\}$. Note that $\bar{R}^S$ is a subring of S . This $\bar{R}^S$ is called the *integral closure* of R in S .

Definition 1.33. R is said to be *integrally closed* in S if $\bar{R}^S = R$.

$$R \hookrightarrow \bar{R}^S \hookrightarrow S.$$

Observation 1.34. $\bar{R}^S$ is integrally closed in S .

Definition 1.35. Let R be an integral domain with quotient field $F = \mathbb{Q}(R)$. We say that R is *integrally closed* if R is integrally closed in F .

Proposition 1.36. If R is a UFD, then R is integrally closed.

Proof. Let $F = \mathbb{Q}(R)$. Say $r/s \in F$ with $r, s \in R$, $s \neq 0$, is integral over R , and suppose r, s have no common prime factors. Then

$$\left(\frac{r}{s}\right)^n + a_{n-1}\left(\frac{r}{s}\right)^{n-1} + \cdots + a_1\left(\frac{r}{s}\right) + a_0 = 0, \quad a_i \in R,$$

so

$$r^n + a_{n-1}r^{n-1}s + \cdots + a_1rs^{n-1} + a_0s^n = 0.$$

Since s divides every term but the first, $s \mid r^n$. FTSOC, if s is not a unit, it has a prime factor p ; then $p \mid r^n$, and since p is prime, $p \mid r$ – contradicting that r, s have no common prime factors. Hence s is a unit in R , so $r/s \in R$. $\square$

Example 1.37. $\mathbb{Z}[\sqrt{-5}]$ is integrally closed but not a UFD. (Exercise.)

Example 1.38. $\mathbb{Z}[\sqrt{5}]$ is an integral domain which is not integrally closed.

Proof. $\mathbb{Q}(\mathbb{Z}[\sqrt{5}]) = \mathbb{Q}(\sqrt{5})$, and

$$\frac{1 + \sqrt{5}}{2} \in \mathbb{Q}(\sqrt{5}) \setminus \mathbb{Z}[\sqrt{5}]$$

is a root of the monic polynomial $X^2 - X - 1 \in \mathbb{Z}[X]$, hence integral over $\mathbb{Z}[\sqrt{5}]$, yet not itself an element of $\mathbb{Z}[\sqrt{5}]$. $\square$

Extensions and Contractions

Let $\varphi : R \rightarrow S$ be a ring morphism. Then φ factors as

$$R \xrightarrow{\varphi} \varphi(R) \xhookrightarrow{i} S \quad \text{with} \quad R / \text{Ker } \varphi \cong \varphi(R).$$

- (1) If $I \subseteq R$ is an ideal, $\varphi(I)$ need *not* be an ideal of S .
- (2) If $J \subseteq S$ is an ideal, $\varphi^{-1}(J)$ is an ideal of R .
- (3) If $Q \in \text{Spec } S$, then $\varphi^{-1}(Q) \in \text{Spec } R$.

Question. Suppose $P \in \text{Spec } R$. Can we say that $\langle \varphi(P) \rangle$ is prime in S ?

Consider $\mathbb{Z} \hookrightarrow \mathbb{Z}[i]: \langle 2 \rangle \mapsto \langle 1+i \rangle \langle 1-i \rangle$. If $p \equiv 1 \pmod{4}$ is prime, then $\langle p \rangle$ is *not* prime in $\mathbb{Z}[i]$ (so the answer to the question above is no, in general).

Exercise 1.39. If $p \equiv 3 \pmod{4}$, then $\langle p \rangle$ is a prime ideal in $\mathbb{Z}[i]$.

Definition 1.40. Let $\varphi : R \rightarrow S$ be a ring morphism.

- For I an ideal of R , the *extension* I^e is the ideal of S generated by $\varphi(I)$.
- For J an ideal of S , the *contraction* $J^c := \varphi^{-1}(J)$, an ideal of R .

Exercise 1.41.

$$I \subseteq I^{ec}, \quad J \supseteq J^{ce}, \quad I^e = I^{ece}, \quad J^c = J^{cec}.$$

Rings of Fractions

Definition 1.42. Let R be a ring. A subset $S \subseteq R$ is called *multiplicative* if

- (1) $1 \in S$,
- (2) $a, b \in S \implies ab \in S$.

Construction 1.43. On $R \times S$, define the relation $\sim$ by

$$(a, s) \sim (b, t) \iff \exists u \in S \text{ such that } u(at - bs) = 0.$$

$\sim$ is transitive. Suppose $(a, s) \sim (b, t)$ and $(b, t) \sim (c, u)$; so $(at - bs)v = 0$ for some $v \in S$, and $(bu - ct)w = 0$ for some $w \in S$. Then

$$atv - bsv = 0 \rightsquigarrow uw \cdot atv - bsuw = 0, \quad buw - ctw = 0 \rightsquigarrow buwsv - ctwsv = 0.$$

Adding,

$$(au - cs)vtw = 0, \quad vtw \in S,$$

so $(a, s) \sim (c, u)$. □

Notation. The equivalence class of (a, s) is denoted $\frac{a}{s}$.

Let

$$S^{-1}R := \{\text{all equivalence classes}\} = \left\{ \frac{a}{s} \mid a \in R, s \in S \setminus \{0\} \right\}.$$

Define

$$\frac{a}{s} + \frac{b}{t} := \frac{at + bs}{st} \quad (\text{well-defined}), \quad \frac{a}{s} \cdot \frac{b}{t} := \frac{ab}{st} \quad (\text{well-defined}).$$

$S^{-1}R$ is a commutative ring with 1. We have a canonical ring morphism

$$\varphi : R \rightarrow S^{-1}R, \quad r \mapsto \frac{r}{1}.$$

Remark 1.44. φ is not injective in general.

Proposition 1.45. Let $g : R \rightarrow A$ be a ring morphism such that $g(S) \subseteq A^\times$ (the units of A). Then there is a unique morphism $h : S^{-1}R \rightarrow A$ such that

$$\begin{array}{ccc} R & \xrightarrow{g} & A \\ \varphi \downarrow & \nearrow h & \\ S^{-1}R & & \end{array}$$

commutes, i.e. $h \circ \varphi = g$.

Proof. **Uniqueness.** Suppose such an h exists. Then

$$h\left(\frac{r}{1}\right) = h(\varphi(r)) = g(r), \quad h\left(\frac{1}{s}\right) = h\left(\left(\frac{s}{1}\right)^{-1}\right) = h\left(\frac{s}{1}\right)^{-1} = g(s)^{-1},$$

so

$$h\left(\frac{r}{s}\right) = h\left(\frac{r}{1}\right)h\left(\frac{1}{s}\right) = g(r)g(s)^{-1},$$

i.e. h is uniquely determined by g .

Existence. Define $h : S^{-1}R \rightarrow A$ by $h(r/s) = g(r)g(s)^{-1}$.

Well-definedness. Suppose $r/s = r'/s'$; then there is $t \in S$ with $t(rs' - r's) = 0$, so

$$g(t)(g(r)g(s') - g(r')g(s)) = 0.$$

Since $g(t) \in A^\times$, $g(r)g(s') = g(r')g(s)$, i.e. $g(r)g(s)^{-1} = g(r')g(s')^{-1}$. □

Example 1.46. Let $P \in \text{Spec } R$ and $S = R \setminus P$. Then S is multiplicative; write $S^{-1}R =: R_P$. Let

$$PR_P := \left\{ \frac{p}{s} \mid p \in P, s \in R \setminus P \right\},$$

an ideal of R_P . If $b/t \notin PR_P$, then $b \notin P$, so $b \in R \setminus P$, so b/t is a unit in R_P . Hence PR_P is the unique maximal ideal of R_P , so R_P is local.

Example 1.47. Let $f \in R$ and $S = \{1, f, f^2, \dots\}$. Then $R_f := S^{-1}R$.

Construction 1.48. Let R be a ring, S a multiplicative subset of R , and M an R -module. On $M \times S$, define $\sim$ by

$$(m, s) \sim (m', s') \iff \exists t \in S \text{ such that } t(ms' - m's) = 0.$$

Then $S^{-1}M := (M \times S)/\sim$ is an $S^{-1}R$ -module.

Let M, N be R -modules and $\varphi : M \rightarrow N$ an R -module morphism. Define $S^{-1}\varphi : S^{-1}M \rightarrow S^{-1}N$ by

$$\frac{m}{s} \mapsto \frac{\varphi(m)}{s}.$$

$S^{-1}\varphi$ is a well-defined $S^{-1}R$ -module morphism.

Proposition 1.49. $S^{-1}(-)$ preserves exactness: if

$$M' \xrightarrow{\varphi} M \xrightarrow{\psi} M''$$

is exact at M , then

$$S^{-1}M' \xrightarrow{S^{-1}\varphi} S^{-1}M \xrightarrow{S^{-1}\psi} S^{-1}M''$$

is exact at $S^{-1}M$.

Proof. We have $\psi \circ \varphi = 0$, so $(S^{-1}\psi) \circ (S^{-1}\varphi) = S^{-1}(\psi \circ \varphi) = 0$. Hence $\text{Im}(S^{-1}\varphi) \subseteq \text{Ker}(S^{-1}\psi)$.

Conversely, let $m/s \in \text{Ker}(S^{-1}\psi)$. Then $\psi(m)/s = 0$, so there is $t \in S$ with $t\psi(m) = 0$, i.e.

$\psi(tm) = 0$, i.e. $tm \in \text{Ker}(\psi) = \text{Im}(\varphi)$. So there is $m' \in M'$ with $\varphi(m') = tm$. Then

$$\frac{m}{s} = \frac{mt}{st} = \frac{\varphi(m')}{st} = (S^{-1}\varphi)\left(\frac{m'}{st}\right) \in \text{Im}(S^{-1}\varphi).$$

Hence $\text{Ker}(S^{-1}\psi) \subseteq \text{Im}(S^{-1}\varphi)$, giving equality. $\square$

1.6 Lecture 6

Theorem 1.50. Let M be an R -module generated by n elements, and let

$$\varphi \in \text{Hom}(M, M) := \{f : M \rightarrow M \mid f \text{ is an } R\text{-module morphism}\}.$$

Let I be an ideal of R such that $\varphi(M) \subseteq IM$. Then there is a relation of the form

$$\varphi^n + a_{n-1}\varphi^{n-1} + \cdots + a_0 = 0, \quad a_i \in I^i.$$

Proof. Let $M = Rw_1 + \cdots + Rw_n$, so $IM = Iw_1 + \cdots + Iw_n$; in particular $\varphi(M) \subseteq IM$. Write

$$\varphi(w_i) = \sum_{j=1}^n a_{ij}w_j, \quad a_{ij} \in I \text{ for all } i,$$

i.e.

$$\sum_{j=1}^n (\varphi \delta_{ij} - a_{ij})w_j = 0.$$

The coefficient matrix has (i, j) -entry $\varphi \delta_{ij} - a_{ij}$, with entries in the commutative subring of $\text{End}(M)$ generated by R (acting by scalars) and φ . Let $d = \det(\text{coefficient matrix})$. Multiplying by the adjugate matrix,

$$d \cdot w_k = 0 \quad \forall k \in [n],$$

so $d \cdot M = 0$, i.e. d is the zero endomorphism of M . Expanding the determinant gives exactly a relation of the stated form. $\square$

Theorem 1.51: Nakayama's Lemma. Let M be a finitely generated R -module and I an ideal of R . If $M = IM$, then there exists $a \in R$ such that $aM = 0$ and $a \equiv 1 \pmod{I}$. Furthermore, if I is the Jacobson radical of R , then $M = 0$.

Proof. Take $\varphi = \text{id}_M$. By the previous theorem, we have

$$1 + a_{n-1} + \cdots + a_0 = 0 \quad (\text{as endomorphisms of } M).$$

Define $a := 1 + a_{n-1} + \cdots + a_0$. Then $aM = 0$ and $a \equiv 1 \pmod{I}$.

If $I \subseteq J(R)$ (the Jacobson radical), then $a - 1 \in J(R)$, so a is a unit; hence $aM = 0$ forces $M = 0$.

$$a^{-1}(aM) = 0. \quad \square$$

Theorem 1.52. Let M be a finitely generated R -module and I an ideal of R such that $M = IM$. Then there exists $i \in I$ such that $(1 + i)M = 0$.

Proof (Ojanguren). Induct on the minimal number of generators $\nu(M) = n$ of M .

If $\nu(M) = 0$, then $M = 0$ and we are done.

Let $M = Rm_1 + \cdots + Rm_n$, and set $M' := M/Rm_n$. Then

$$IM' = \frac{IM + Rm_n}{Rm_n} = \frac{M + Rm_n}{Rm_n} = \frac{M}{Rm_n} = M',$$

using $IM = M$. Since $\nu(M') < n$, by induction there is $x \in I$ such that $(1 + x)M' = 0$, i.e.

$$(1 + x) \frac{M}{Rm_n} = 0 \implies (1 + x)M \subseteq Rm_n.$$

Then

$$(1 + x)M = (1 + x)IM = I(1 + x)M \subseteq I \cdot Rm_n = Im_n,$$

using $(1 + x)M \subseteq Rm_n$ in the last containment. In particular $(1 + x)m_n \in Im_n$, so $(1 + x)m_n = ym_n$ for some $y \in I$, giving

$$(1 + x - y)m_n = 0.$$

Now for any $\mu \in M$, $(1 + x)\mu = rm_n$ for some $r \in R$ (since $(1 + x)M \subseteq Rm_n$), so

$$(1 + x - y)(1 + x)\mu = r(1 + x - y)m_n = 0.$$

Hence $(1 + x)(1 + x - y)M = 0$. Expanding, $(1 + x)(1 + x - y) = 1 + i$ where $i := x + (x - y) + x(x - y) \in I$. So $(1 + i)M = 0$. $\square$

Definition 1.53. Let M, N be R -modules. The *tensor product* of M, N is a pair (T, θ) such that T is an R -module, $\theta : M \times N \rightarrow T$ is bilinear, and for every bilinear map $f : M \times N \rightarrow P$, there is a unique linear map $h : T \rightarrow P$ such that

$$\begin{array}{ccc} M \times N & \xrightarrow{f} & P \\ \theta \downarrow & \nearrow h & \\ T & & \end{array}$$

commutes, i.e. $h \circ \theta = f$.

Recall the canonical morphism $R \rightarrow S^{-1}R$, $r \mapsto r/1$, and let I be an ideal of R . An element of the

extended ideal I^e is a finite sum

$$\frac{r_1}{s_1} \cdot \frac{x_1}{1} + \cdots + \frac{r_n}{s_n} \cdot \frac{x_n}{1} = \frac{s'_1 r_1 x_1 + \cdots + s'_n r_n x_n}{s_1 \cdots s_n},$$

where $r_i \in R$, $s_i \in S$, $x_i \in I$ (writing s'_i for the product of all s_j with $j \neq i$).

Proposition 1.54.

- (1) Every ideal of $S^{-1}R$ is an extended ideal.
- (2) The prime ideals of $S^{-1}R$ are in one-to-one correspondence, $P \leftrightarrow S^{-1}P$, with the prime ideals P of R such that $S \cap P = \emptyset$.

Proof. See [AM69]. □

Local Properties of Modules

Proposition 1.55. Let M be an R -module. Then the following are equivalent:

- (1) $M = 0$;
- (2) $M_P = 0$ for all $P \in \text{Spec}(R)$;
- (3) $M_{\mathfrak{m}} = 0$ for all $\mathfrak{m} \in \text{Max}(R)$.

(Here for $P \in \text{Spec}(R)$, writing $S = R \setminus P$, we set $M_P := S^{-1}M$.)

Proof. (1) $\Rightarrow$ (2) $\Rightarrow$ (3) trivially.

(3) $\Rightarrow$ (1): Let $M \neq 0$; then there is $x \in M$ with $x \neq 0$. Let $I = \text{Ann}(x) = \{r \in R \mid rx = 0\}$, a proper ideal of R . Let $\mathfrak{m}$ be a maximal ideal with $I \subseteq \mathfrak{m}$.

Consider $x/1 \in M_{\mathfrak{m}}$. Since $M_{\mathfrak{m}} = 0$, $x/1 = 0$, so there is $t \in R \setminus \mathfrak{m}$ with $tx = 0$. Then $t \in I \subseteq \mathfrak{m}$ – contradicting $t \notin \mathfrak{m}$. □

Exercise 1.56. Let $\varphi \in \text{Hom}(M, N)$. TFAE:

- (1) φ is injective (resp. surjective);
- (2) $\varphi_P : M_P \rightarrow N_P$ is injective (resp. surjective) for all $P \in \text{Spec}(R)$;
- (3) $\varphi_{\mathfrak{m}} : M_{\mathfrak{m}} \rightarrow N_{\mathfrak{m}}$ is injective (resp. surjective) for all $\mathfrak{m} \in \text{Max}(R)$.

Proposition 1.57. Let $R \hookrightarrow S$ be rings with S integral over R .

- (1) If J is an ideal of S , then $R/(J \cap R) \hookrightarrow S/J$ is integral.
- (2) If T is a multiplicative subset of R , then $T^{-1}R \hookrightarrow T^{-1}S$ is integral.

Proof. (1) is straightforward (reduce the monic integral equation modulo J).

(2) Let $x/t \in T^{-1}S$ ($x \in S, t \in T$). Since x is integral over R ,

$$x^n + a_1x^{n-1} + \cdots + a_n = 0 \quad \text{for some } a_i \in R.$$

Dividing by t^n ,

$$\left(\frac{x}{t}\right)^n + \frac{a_1}{t} \left(\frac{x}{t}\right)^{n-1} + \cdots + \frac{a_n}{t^n} = 0,$$

so x/t is integral over $T^{-1}R$. □

Proposition 1.58. Let $R \hookrightarrow S$ be an integral extension, with R, S both integral domains. Then R is a field if and only if S is a field.

Proof. ($\Rightarrow$) Let R be a field and $y \in S \setminus \{0\}$. Since y is integral over R , take a monic relation of minimal degree,

$$y^n + a_1y^{n-1} + \cdots + a_n = 0, \quad a_i \in R.$$

If $a_n = 0$, then $y(y^{n-1} + a_1y^{n-2} + \cdots + a_{n-1}) = 0$; since S is a domain and $y \neq 0$, this gives a relation of degree $n - 1$, contradicting minimality. So $a_n \neq 0$, and as R is a field, a_n is a unit. Then

$$y(y^{n-1} + \cdots + a_{n-1}) = -a_n,$$

and since $-a_n$ is a unit, y is a unit.

($\Leftarrow$) Let S be a field. Take any $x \in R \setminus \{0\}$. Then $x^{-1} \in S$ is integral over R :

$$x^{-m} + a_1x^{-(m-1)} + \cdots + a_m = 0, \quad a_i \in R,$$

so

$$x^{-1} = -(a_1 + a_2x + \cdots + a_mx^{m-1}) \in R.$$

□

1.7 Lecture 7

Example 1.59. Let k be a field, t an indeterminate, and $A = k[t^2, t^3] \subseteq B = k[t]$. Since A is a subring of the integral domain B , A is an integral domain. The fraction field of both A and B is $k(t)$, and since B is a UFD, B is integrally closed in $k(t)$.

Question. Is A integrally closed in $k(t)$?

No: t is integral over A , being a root of $X^2 - t^2 \in A[X]$, but $t \notin A$. So A is not integrally closed.

Question. What is the integral closure of A in $k(t)$?

It is B , since B is integrally closed in $k(t)$.

Example 1.60. Similarly,

$$\begin{array}{ccc} B & \hookrightarrow & \mathbb{Q}(i) \\ \uparrow & & \uparrow \\ \mathbb{Z} & \hookrightarrow & \mathbb{Q} \end{array}$$

where B is the integral closure of $\mathbb{Z}$ (an integrally closed domain) in $\mathbb{Q}(i)$. One checks $B = \mathbb{Z}[i]$: $\min(i, \mathbb{Q}) = X^2 + 1 \in \mathbb{Z}[X]$.

Theorem 1.61. Let A be an integrally closed domain with $K = \mathbb{Q}(A)$, and let L be an algebraic extension of K . Then $\alpha \in L$ is integral over A if and only if $\min(\alpha, K) \in A[X]$.

Proof. ($\Leftarrow$) is trivial: if $f(X) = X^n + a_1X^{n-1} + \cdots + a_n = \min(\alpha, K) \in A[X]$, then $f(\alpha) = 0$ exhibits α as integral over A .

($\Rightarrow$) Let α be integral over A ; we show the coefficients a_i of $f := \min(\alpha, K)$ lie in A .

Let $\bar{L}$ be an algebraic closure of L . In $\bar{L}$, f splits as

$$f(X) = (X - \alpha_1) \cdots (X - \alpha_n),$$

where the α_i are the conjugates of α over K . For each i there is a K -isomorphism $K[\alpha] \cong K[\alpha_i]$ fixing K and sending $\alpha \mapsto \alpha_i$.

Since α is integral over A , there is a monic $g(X) \in A[X]$ with $g(\alpha) = 0$; as $f = \min(\alpha, K)$, $f \mid g$ in $K[X]$. Applying the isomorphism above, $g(\alpha_i) = 0$ for each i , so each α_i is integral over A .

Now $a_1, \dots, a_n \in A[\alpha_1, \dots, \alpha_n]$ (they are, up to sign, the elementary symmetric functions of the α_i – Newton's identities), and each α_i is integral over A , so each a_i is integral over A . But $a_i \in K$ and A is integrally closed in K , so $a_i \in A$ for all i . $\square$

Exercise 1.62.

- (1) $k[X, Y] / \langle Y^2 - X^3 \rangle \cong k[t^2, t^3]$ (the coordinate ring of the curve $Y^2 = X^3$).
- (2) Show directly that $k[t^2, t^3]$ is not a UFD.

Example 1.63. Let A be a UFD with $2 \in A^\times$, and let $f \in A$ be square-free. Then $A[\sqrt{f}]$ is an integrally closed domain.

Proof. Since $\langle X^2 - f \rangle \in \text{Spec}(A[X])$ (as f is square-free, $X^2 - f$ is irreducible),

$$A[\sqrt{f}] \cong A[X] / \langle X^2 - f \rangle$$

is an integral domain.

Let α be a square root of f ; we ask whether $A[\alpha]$ is integrally closed. Let $K := \mathbb{Q}(A)$; since A is a

UFD, A is integrally closed in K .

Case $\alpha \in K$: Then $X^2 - f \in A[X]$ has root $\alpha \in K$, so α is integral over A , hence $\alpha \in A$ (as A is integrally closed in K). So $A[\alpha] = A$, trivially integrally closed.

Case $\alpha \notin K$: Then

$$\mathbb{Q}(A[\alpha]) = K(\alpha) = K + K\alpha,$$

and every $\theta \in K(\alpha)$ can be written uniquely as $\theta = x + y\alpha$ with $x, y \in K$; its minimal polynomial over K is

$$X^2 - 2xX + (x^2 - fy^2).$$

Suppose θ is integral over $A[\alpha]$. Since α is integral over A (a root of $X^2 - f \in A[X]$), $A[\alpha]$ is integral over A ; by transitivity, θ is integral over A . By the previous theorem (as A is integrally closed in K),

$$X^2 - 2xX + (x^2 - fy^2) \in A[X],$$

i.e. $2x \in A$ and $x^2 - fy^2 \in A$. Since 2 is a unit, $x \in A$; hence $fy^2 = x^2 - (x^2 - fy^2) \in A$.

If some prime $p \in A$ divides the denominator of y (writing $y \in K$ in lowest terms), then since $fy^2 \in A$ and f is square-free, we would need $p^2 \mid f$ – a contradiction. Hence $y \in A$, so $\theta = x + y\alpha \in A[\alpha]$.

Together with the trivial case $A[\alpha] = A$, this shows $A[\alpha]$ is integrally closed. $\square$

Lemma 1.64. Let $A \hookrightarrow B$ be an integral extension. If $P \in \text{Max}(B)$, then $P \cap A \in \text{Max}(A)$. Conversely, if $\mathfrak{p} \in \text{Max}(A)$, then there exists $P \in \text{Spec}(B)$ such that $P \cap A = \mathfrak{p}$ (and P is maximal too).

Proof. $P \cap A \in \text{Spec}(A)$, and $A/(P \cap A) \hookrightarrow B/P$ is an integral extension. Since B/P is a field, $A/(P \cap A)$ is a field, so $P \cap A \in \text{Max}(A)$.

Conversely, let $\mathfrak{p} \in \text{Max}(A)$. It suffices to show $\mathfrak{p}B \neq B$: if $\mathfrak{p}B$ is a proper ideal, there is $P \in \text{Max}(B)$ with $\mathfrak{p}B \subseteq P$, so $\mathfrak{p} \subseteq P \cap A$; since $1 \notin P \cap A$ and $\mathfrak{p}$ is maximal, $P \cap A = \mathfrak{p}$.

Suppose instead $\mathfrak{p}B = B$. Then $1 = b_1p_1 + \cdots + b_np_n$ for some $b_i \in B, p_i \in \mathfrak{p}$. Let $C = A[b_1, \dots, b_n]$; since each b_i is integral over A , C is a finitely generated A -module. From $1 = \sum b_ip_i$ (with $b_i \in C, p_i \in \mathfrak{p}$), $\mathfrak{p}C = C$.

By Nakayama's Lemma, there is $a \in A$ with $a \equiv 1 \pmod{\mathfrak{p}}$ and $aC = 0$; since $1 \in C, a = a \cdot 1 = 0$, so $0 \equiv 1 \pmod{\mathfrak{p}}$, i.e. $1 \in \mathfrak{p}$ – a contradiction. $\square$

Theorem 1.65: Lying Over. Let $A \hookrightarrow B$ be an integral extension and $\mathfrak{p} \in \text{Spec}(A)$.

- (i) There exists $P \in \text{Spec}(B)$ such that $P \cap A = \mathfrak{p}$ (we say P lies over $\mathfrak{p}$).
- (ii) There is no inclusion between distinct prime ideals of B lying over $\mathfrak{p}$: if $P \subseteq Q$ and $P \cap A = \mathfrak{p} = Q \cap A$, then $P = Q$.

Proof. Localize at $S = A \setminus \mathfrak{p}$: we obtain $A_{\mathfrak{p}} \hookrightarrow B_{\mathfrak{p}} := S^{-1}B$, an integral extension (by the localization-

integral-extension theorem below), with $A_{\mathfrak{p}}$ local with maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. By the previous lemma (applied to $A_{\mathfrak{p}} \hookrightarrow B_{\mathfrak{p}}$ and the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$), there is a maximal ideal $\mathfrak{P}$ of $B_{\mathfrak{p}}$ with $\mathfrak{P} \cap A_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$.

Let $P \subseteq B$ be the contraction of $\mathfrak{P}$ along $B \rightarrow B_{\mathfrak{p}}$. Then $P \cap A = \mathfrak{p}$, proving (i).

Moreover, the primes of B lying over $\mathfrak{p}$ are in bijective correspondence with the maximal ideals of $B_{\mathfrak{p}}$ lying over $\mathfrak{p}A_{\mathfrak{p}}$ (via this same contraction/extension correspondence). Since distinct maximal ideals are never comparable by inclusion, neither are the corresponding primes of B lying over $\mathfrak{p}$, proving (ii). $\square$

Theorem 1.66. Let $A \hookrightarrow B$ be an extension of rings and C the integral closure of A in B . If S is a multiplicative subset of A , then $S^{-1}C$ is the integral closure of $S^{-1}A$ in $S^{-1}B$.

Proof. Let $b/s \in S^{-1}B$ be integral over $S^{-1}A$:

$$\left(\frac{b}{s}\right)^n + \frac{a_1}{s_1} \left(\frac{b}{s}\right)^{n-1} + \cdots + \frac{a_n}{s_n} = 0.$$

Let $t = s_1 \cdots s_n$. Multiplying through by $(st)^n$,

$$(st)^n \left[\left(\frac{b}{s}\right)^n + \cdots \right] = 0$$

becomes a monic equation in bt with coefficients in A . Hence bt is integral over A , i.e. $bt \in C$, so

$$\frac{b}{s} = \frac{bt}{st} \in S^{-1}C.$$

$\square$

1.8 Lecture 8

1.9 Lecture 9

Proposition 1.67. Let $\varphi : A \rightarrow B$ be a ring morphism and $\mathfrak{p} \in \text{Spec } A$. Then $\mathfrak{p}$ is the contraction of some $Q \in \text{Spec } B$ if and only if $\mathfrak{p}^{ec} = \mathfrak{p}$.

Proof. In general, $\mathfrak{p} \subseteq \mathfrak{p}^{ec}$.

($\Rightarrow$) If $\mathfrak{p} = Q^c$ for some $Q \in \text{Spec } B$, then

$$\mathfrak{p}^e = Q^{ce} \subseteq Q \implies \mathfrak{p}^{ec} \subseteq Q^c = \mathfrak{p}.$$

($\Leftarrow$) Assume $\mathfrak{p}^{ec} = \mathfrak{p}$. Let $S = \varphi(A \setminus \mathfrak{p})$, a multiplicative subset of B .

Note: $\mathfrak{p}^e \cap S = \emptyset$. Indeed, if $x \in \mathfrak{p}^e \cap S$, then $\varphi^{-1}(x) \in A \setminus \mathfrak{p}$; but $x \in \mathfrak{p}^e$ gives $\varphi^{-1}(x) \in \mathfrak{p}^{ec} = \mathfrak{p}$ – a contradiction.

So $\mathfrak{p}^e S^{-1}B$ is a proper ideal of $S^{-1}B$, hence contained in some $\mathfrak{m} \in \text{Max}(S^{-1}B)$. Let Q be the contraction of $\mathfrak{m}$ in B ; then $Q \in \text{Spec } B$ and $\mathfrak{p}^e \subseteq Q$. Moreover $Q \cap S = \emptyset$, which gives $Q^c \subseteq \mathfrak{p}$.

Now $\mathfrak{p}^{ec} \subseteq Q^c \subseteq \mathfrak{p}$, and since $\mathfrak{p}^{ec} = \mathfrak{p}$, we get $\mathfrak{p} = Q^c$. $\square$

Proposition 1.68. Let $A \hookrightarrow B$ be rings and C the integral closure of A in B . Let $I \subseteq A$ be an ideal. Then the integral closure of I in B is $\sqrt{IC} \subseteq C$.

Proof. Suppose x is integral over I in B :

$$x^n + a_1 x^{n-1} + \cdots + a_n = 0, \quad a_i \in I.$$

Then $x \in C$ (a fortiori integral over A), and

$$x^n = -(a_1 x^{n-1} + \cdots + a_n) \in IC,$$

so $x \in \sqrt{IC}$.

Conversely, let $x \in \sqrt{IC}$, so $x^n \in IC$ for some $n > 0$:

$$x^n = a_1 x_1 + \cdots + a_m x_m, \quad a_i \in I, x_i \in C.$$

Each $x_i \in C$ is integral over A , so $M := A[x_1, \dots, x_m]$ is a finitely generated A -module, and $x^n M \subseteq IM$.

Consider the endomorphism $\mu_{x^n} : M \rightarrow M$ given by multiplication by x^n ; since $x^n M \subseteq IM$, the determinant-trick argument underlying Nakayama's Lemma (Lecture 6) yields a monic relation

$$(x^n)^k + c_{k-1}(x^n)^{k-1} + \cdots + c_0 = 0, \quad c_i \in I^i,$$

as endomorphisms of M . Evaluating at $1 \in M$ (since $1 \in A \subseteq M$) gives this same relation as an identity in B , exhibiting x as integral over I . $\square$

Proposition 1.69. Let $A \hookrightarrow B$ be integral domains, $I \subseteq A$ an ideal, and suppose A is integrally closed with $Q(A) = K$. Let $x \in B$ be integral over I . Then x is algebraic over K , and if

$$\min(x, K) = T^n + a_1 T^{n-1} + \cdots + a_n,$$

then $a_i \in \sqrt{I}$ for all i .

Proof. Clearly x is algebraic over K (being integral over $A \subseteq K$). Let L be an algebraic extension of K containing the conjugates $x_1, \dots, x_n$ of x .

Each x_i is integral over I (as in the proof of the minimal-polynomial theorem from Lecture 7). Using the relations between roots and coefficients, $a_1, \dots, a_n$ are all integral over I .

Applying the previous proposition with B replaced by K (in which the integral closure of A is A itself, since A is integrally closed), we get $a_i \in \sqrt{I}$ for all i . $\square$

Theorem 1.70: Going Down. Assume:

- (a) $A \hookrightarrow B$ is an integral extension;
- (b) A, B are integral domains;
- (c) A is integrally closed.

Let $\mathfrak{p}_1 \supseteq \mathfrak{p}_2 \supseteq \cdots \supseteq \mathfrak{p}_n$ be a chain in $\text{Spec } A$, and let $Q_1 \supseteq \cdots \supseteq Q_m$ ($m \leq n$) be a chain in $\text{Spec } B$ with $Q_i \cap A = \mathfrak{p}_i$ for all $i \in [m]$. Then the chain can be extended:

$$Q_1 \supseteq \cdots \supseteq Q_m \supseteq Q_{m+1} \supseteq \cdots \supseteq Q_n$$

with $Q_i \in \text{Spec } B$ and $Q_i \cap A = \mathfrak{p}_i$ for all $i \in [n]$.

Proof. It suffices to treat $m = 1, n = 2$: we need $Q_2 \in \text{Spec } B$ with $Q_1 \supseteq Q_2$ and $Q_2 \cap A = \mathfrak{p}_2$.

The prime ideals of B_{Q_1} are in bijective correspondence with the primes of B contained in Q_1 , via

$$A \hookrightarrow B \hookrightarrow B_{Q_1}.$$

It suffices to show $\mathfrak{p}_2 B_{Q_1} \cap A = \mathfrak{p}_2$. The containment $\mathfrak{p}_2 \subseteq \mathfrak{p}_2 B_{Q_1} \cap A$ is clear; we show the reverse.

Let $x \in \mathfrak{p}_2 B_{Q_1} \cap A$, so $x = y/s$ with $y \in \mathfrak{p}_2 B, s \in B \setminus Q_1$. Write

$$y = r_1 b_1 + \cdots + r_k b_k, \quad r_i \in \mathfrak{p}_2, b_i \in B.$$

Since B is integral over A , each b_i is integral over A , so y is integral over $\mathfrak{p}_2$ (by the previous propositions).

By the proposition on minimal polynomials (applied with $I = \mathfrak{p}_2$), $\min(y, K)$ (where $K = \mathbb{Q}(A)$) has all coefficients in $\sqrt{\mathfrak{p}_2} = \mathfrak{p}_2$:

$$y^r + u_1 y^{r-1} + \cdots + u_r = 0, \quad u_i \in \mathfrak{p}_2.$$

Since $x = y/s \in A$, we have $s = yx^{-1}$ with $x^{-1} \in K$ (the case $x = 0$ being trivial). Substituting $y = sx$ into the relation above and dividing by x^r ,

$$s^r + \frac{u_1}{x} s^{r-1} + \cdots + \frac{u_r}{x^r} = 0,$$

a minimal equation for s over K . Since $s \in B$ is integral over A , applying the proposition again (with $I = A$) gives $u_i/x^i \in \sqrt{A} = A$ for all i .

Let $v_i := u_i/x^i \in A$, so $v_i x^i = u_i \in \mathfrak{p}_2$. FTSOC, suppose $x \notin \mathfrak{p}_2$. Since $\mathfrak{p}_2$ is prime, $x^i \notin \mathfrak{p}_2$, so $v_i \in \mathfrak{p}_2$ for all i . Then

$$s^r = - \left(\frac{u_1}{x} s^{r-1} + \cdots + \frac{u_r}{x^r} \right) \in \mathfrak{p}_2 B \subseteq \mathfrak{p}_1 B \subseteq Q_1,$$

so $s^r \in Q_1$; since Q_1 is prime, $s \in Q_1$ – contradicting $s \in B \setminus Q_1$.

Hence $x \in \mathfrak{p}_2$, completing the proof. $\square$

Tensor Products of Modules

Definition 1.71. Let R be a commutative ring and M, N, P be R -modules. A *tensor product* of M, N is a pair (T, θ) , where T is an R -module and $\theta : M \times N \rightarrow T$ is bilinear, such that for every bilinear map $f : M \times N \rightarrow P$, there is a unique linear map $h : T \rightarrow P$ with

$$\begin{array}{ccc} M \times N & \xrightarrow{f} & P \\ \theta \downarrow & \nearrow h & \\ T & & \end{array}$$

i.e. $h \circ \theta = f$.

Exercise 1.72. The tensor product, if it exists, is unique up to isomorphism.

Remark 1.73. $\text{Im}(\theta)$ generates T .

Proof. Let T' be the submodule of T generated by $\text{Im}(\theta)$, with inclusion $i : T' \hookrightarrow T$. Since $\text{Im}(\theta) \subseteq T'$, θ corestricts to a bilinear map $\theta' : M \times N \rightarrow T'$. By the universal property of (T, θ) applied to the bilinear map θ' (viewed as landing in T via $i \circ \theta' = \theta$), there is a unique $h : T \rightarrow T'$ with $h \circ \theta = \theta'$; composing with i , $(i \circ h) \circ \theta = i \circ \theta' = \theta$.

But id_T also satisfies $\text{id}_T \circ \theta = \theta$; by the uniqueness clause of the universal property, $i \circ h = \text{id}_T$. Hence i is surjective, and since i is also injective (being an inclusion), $T' = T$. $\square$

Construction 1.74. Let (F, ι) be the free R -module on the set $M \times N$. Let H be the submodule of F generated by all elements of the form

- (a) $\iota(m_1 + m_2, n) - \iota(m_1, n) - \iota(m_2, n)$,
- (b) $\iota(m, n_1 + n_2) - \iota(m, n_1) - \iota(m, n_2)$,
- (c) $\iota(rm, n) - r \iota(m, n)$,

for $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, $r \in R$. Write $M \otimes_R N := F/H$, and let $\pi : F \rightarrow F/H$ be the natural projection.

$\pi \circ \iota$ is bilinear. For instance,

$$(\pi \circ \iota)(m_1 + m_2, n) = \iota(m_1 + m_2, n) + H = (\iota(m_1, n) + \iota(m_2, n)) + H = (\pi \circ \iota)(m_1, n) + (\pi \circ \iota)(m_2, n),$$

using relation (a); the other bilinearity identities follow similarly from (b) and (c). $\square$

Write $\otimes_R := \pi \circ \iota : M \times N \rightarrow M \otimes_R N$, so that $\otimes_R(m, n) =: m \otimes n$ are the *simple tensors*.

$(M \otimes_R N, \otimes_R)$ satisfies the universal property. Let P be an R -module and $g : M \times N \rightarrow P$ bilinear. Since F is free on $M \times N$, there is a unique morphism $j : F \rightarrow P$ such that $j \circ \iota = g$.

$H \subseteq \ker j$. For instance,

$$j(\iota(m_1 + m_2, n) - \iota(m_1, n) - \iota(m_2, n)) = g(m_1 + m_2, n) - g(m_1, n) - g(m_2, n) = 0,$$

using bilinearity of g ; similarly for the other generators of H .

Hence j induces a morphism $\bar{k} : M \otimes_R N \rightarrow P$ with $\bar{k} \circ \pi = j$, i.e.

$$\bar{k} \circ \otimes_R = \bar{k} \circ (\pi \circ \iota) = (\bar{k} \circ \pi) \circ \iota = j \circ \iota = g.$$

Uniqueness. Let $k : M \otimes_R N \rightarrow P$ be any morphism with $k \circ \otimes_R = g$, i.e. $k \circ \pi \circ \iota = g = \bar{k} \circ \pi \circ \iota$. By the uniqueness property of the free module F , $k \circ \pi = \bar{k} \circ \pi$; since π is onto, $k = \bar{k}$. $\square$

1.10 Lecture 10

Construction 1.75. Define $R \times (R \otimes M) \rightarrow R \otimes M$ by

$$\left(s, \sum_{i=1}^k r_i \otimes m_i \right) \mapsto \sum_{i=1}^k (sr_i) \otimes m_i.$$

One checks this makes $R \otimes M$ an R -module.

Theorem 1.76. $R \otimes_R M \cong M$ as R -modules.

Proof. The map $R \times M \rightarrow M$, $(r, m) \mapsto rm$, is bilinear, so it induces a unique morphism $f : R \otimes M \rightarrow M$ with $f(r \otimes m) = rm$.

Define $g : M \rightarrow R \otimes M$ by $g(m) = 1 \otimes m$. Then f is clearly surjective, and f, g are mutually inverse. $\square$

From the definition of the tensor product,

$$\{\text{bilinear maps } M \times N \rightarrow P\} \longleftrightarrow \{\text{module morphisms } M \otimes N \rightarrow P\} =: \text{Hom}_R(M \otimes N, P).$$

Remark 1.77. If $M_1, \dots, M_n$ are R -modules, $M_1 \otimes M_2 \otimes \dots \otimes M_n$ can be defined in exactly the same way.

Proposition 1.78. Let M, N, P be R -modules. There are unique isomorphisms such that:

- (a) $M \otimes N \cong N \otimes M$;

- (b) $(M \otimes N) \otimes P \cong M \otimes (N \otimes P) \cong M \otimes N \otimes P$, with $(m \otimes n) \otimes p \mapsto m \otimes (n \otimes p) \mapsto m \otimes n \otimes p$;
- (c) $(M \oplus N) \otimes P \cong (M \otimes P) \oplus (N \otimes P)$, with $(m \oplus n) \otimes p \mapsto (m \otimes p) \oplus (n \otimes p)$;
- (d) $R \otimes M \cong M$, with $r \otimes m \mapsto rm$.

Proof. (a), (c), (d) are straightforward verifications via the universal property; (b) is similar. $\square$

Proposition 1.79. Let $x_i \in M, y_i \in N$ be such that $\sum x_i \otimes y_i = 0$ in $M \otimes N$ (a finite sum). Then there exist finitely generated submodules $M' \subseteq M$ and $N' \subseteq N$ such that $\sum x_i \otimes y_i = 0$ already in $M' \otimes N'$.

Proof. Recall $M \otimes N := F/H$, where F is free on $M \times N$. Since $\sum x_i \otimes y_i = 0$, the element $\sum \iota(x_i, y_i) \in H$ is a finite sum of the generators of H .

Let M' be the submodule of M generated by the x_i 's together with the first coordinates appearing in these generators of H , and similarly N' using the y_i 's and second coordinates. These are finitely generated, and the relation exhibiting $\sum \iota(x_i, y_i) \in H$ only involves elements of M' and N' , so $\sum x_i \otimes y_i = 0$ already in $M' \otimes N'$. $\square$

Definition 1.80. An R -algebra is a set S such that:

- (1) S is an R -module;
- (2) S is a ring with 1 (associative);
- (3) $r(s_1 s_2) = (rs_1)s_2 = s_1(rs_2)$ for all $r \in R, s_1, s_2 \in S$.

Example 1.81. Let R, S be rings and $f : R \rightarrow S$ a morphism such that $f(R) \subseteq Z(S)$ (the center of S). Then S is an R -algebra, via

$$R \times S \rightarrow S, \quad (r, s) \mapsto f(r) \cdot s.$$

Example 1.82. Let R, S be commutative rings and $f : R \rightarrow S$ a ring morphism. Then S is an R -algebra (a special case of the above).

Remark 1.83. Every ring is a $\mathbb{Z}$ -algebra, via $f : \mathbb{Z} \rightarrow R, n \mapsto n \cdot 1_R$.

Definition 1.84. Let $f : R \rightarrow S$ be a ring morphism and N an S -module. Define $R \times N \rightarrow N$ by $(r, n) \mapsto f(r) \cdot n$; this makes N an R -module (*restriction of scalars*). In particular, S is an R -algebra.

Definition 1.85. Let M be an R -module and S an R -algebra. Define

$$S \times (M \otimes_R S) \rightarrow M \otimes_R S, \quad (s, m \otimes t) \mapsto m \otimes st.$$

Then $M \otimes_R S$ is an S -module (*extension of scalars*).

Remark 1.86. Let M be a finitely generated R -module, generated by $\{x_1, \dots, x_n\}$. Then $M \otimes_R S$ is a finitely generated S -module, generated by $\{x_1 \otimes 1, \dots, x_n \otimes 1\}$.

Theorem 1.87.

$$\mathrm{Hom}_R(M \otimes N, P) \cong \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P)).$$

Proof. Let $f : M \times N \rightarrow P$ be bilinear. For each $x \in M$, $y \mapsto f(x, y)$ gives a linear map $N \rightarrow P$. Hence

$$x \longmapsto (y \mapsto f(x, y))$$

is a linear map from M to $\mathrm{Hom}_R(N, P)$.

Conversely, take a morphism $\varphi : M \rightarrow \mathrm{Hom}_R(N, P)$. Define $M \times N \rightarrow P$ by $(m, n) \mapsto \varphi(m)(n)$; this is bilinear.

These two constructions are mutually inverse, giving the desired isomorphism. $\square$

Notation. Fact. $M' \rightarrow M \rightarrow M'' \rightarrow 0$ is exact if and only if, for every R -module D ,

$$0 \rightarrow \mathrm{Hom}(M'', D) \rightarrow \mathrm{Hom}(M, D) \rightarrow \mathrm{Hom}(M', D)$$

is exact.

Let $f : M \rightarrow M'$ and $g : N \rightarrow N'$ be morphisms. Define $M \times N \rightarrow M' \otimes N'$ by $(m, n) \mapsto f(m) \otimes g(n)$; this is bilinear, so it induces a morphism

$$f \otimes g : M \otimes N \rightarrow M' \otimes N', \quad (f \otimes g)(m \otimes n) = f(m) \otimes g(n).$$

Proposition 1.88. Let $M' \xrightarrow{f} M \xrightarrow{g} M'' \rightarrow 0$ be exact, and let N be any R -module. Then

$$M' \otimes N \xrightarrow{f \otimes 1} M \otimes N \xrightarrow{g \otimes 1} M'' \otimes N \rightarrow 0$$

is exact.

Proof. Call the first sequence E and the second $E \otimes N$. Let P be an R -module.

E exact implies

$$0 \rightarrow \mathrm{Hom}(M'', \mathrm{Hom}(N, P)) \rightarrow \mathrm{Hom}(M, \mathrm{Hom}(N, P)) \rightarrow \mathrm{Hom}(M', \mathrm{Hom}(N, P))$$

is exact (by the Fact above, applied with $D = \text{Hom}(N, P)$), which (using the Tensor-Hom adjunction) gives

$$0 \rightarrow \text{Hom}(M'' \otimes N, P) \rightarrow \text{Hom}(M \otimes N, P) \rightarrow \text{Hom}(M' \otimes N, P)$$

exact. By the Fact again, $M' \otimes N \rightarrow M \otimes N \rightarrow M'' \otimes N \rightarrow 0$ is exact. $\square$

Theorem 1.89. Let R be a ring, $S \subseteq R$ multiplicative, and M an R -module. Then

$$S^{-1}R \otimes_R M \cong S^{-1}M.$$

Proof. Define $S^{-1}R \times M \rightarrow S^{-1}M$ by $(a/s, m) \mapsto am/s$; this is R -linear. This induces a morphism

$$f : S^{-1}R \otimes_R M \rightarrow S^{-1}M, \quad \frac{a}{s} \otimes m \mapsto \frac{am}{s},$$

uniquely determined, and clearly surjective.

Injectivity. Take a typical element of $S^{-1}R \otimes_R M$,

$$\frac{a_1}{s_1} \otimes m_1 + \cdots + \frac{a_n}{s_n} \otimes m_n.$$

Let $s = s_1 \cdots s_n$ and $t_j = \prod_{i \neq j} s_i$, so $a_j/s_j = a_j t_j/s$. Then

$$\frac{a_j}{s_j} \otimes m_j = \frac{a_j t_j}{s} \otimes m_j = \frac{1}{s} \otimes (a_j t_j m_j),$$

so the element becomes

$$\frac{1}{s} \otimes (a_1 t_1 m_1) + \cdots + \frac{1}{s} \otimes (a_n t_n m_n) = \frac{1}{s} \otimes m, \quad m := a_1 t_1 m_1 + \cdots + a_n t_n m_n.$$

Suppose $f(1/s \otimes m) = 0$, i.e. $m/s = 0$ in $S^{-1}M$. Then there is $t \in S$ with $tm = 0$, and

$$\frac{1}{s} \otimes m = \frac{t}{st} \otimes m = \frac{1}{st} \otimes (tm) = \frac{1}{st} \otimes 0 = 0.$$

Hence f is injective, so $S^{-1}R \otimes_R M \cong S^{-1}M$. $\square$

CHAPTER 2

PROBLEM DISCUSSIONS

2.1 Assignment 1

Convention. By a ring, we mean a commutative ring with 1. Also, we exclude the zero ring.

1. Let F be a field. Prove that $F[X]$ has infinitely many irreducible polynomials.
2. Let R be a ring. Show that $R[X]$ is never a local ring.
3. Let R be a ring in which every element is either nilpotent or a unit. Prove that R is a local ring. How many prime ideals can R have?
4. Let R be a ring. Prove that $a \in J(R)$ if and only if $1 + ab$ is a unit for all $b \in R$. (In particular, $1 + a$ is a unit.)
5. Let R be a ring and I be a finitely generated ideal such that $I^2 = I$. Show that there exists $e \in I$ such that $I = \langle e \rangle$ where $e^2 = e$ (such an element is called idempotent).
6. Let R be a ring and I be a finitely generated ideal such that $I = \langle a_1, \dots, a_n \rangle + I^2$. Show that there exists $e \in I^2$ such that $I = \langle a_1, \dots, a_n, e \rangle$ where $e(1 - e) \in \langle a_1, \dots, a_n \rangle$. What happens if R is a local ring?
7. Is $R[X] \hookrightarrow R[X, X^{-1}]$ an integral extension?
8. Let $A \hookrightarrow B$ be an integral extension where A, B are integral domains. Let C be an integral domain and $\varphi : B \rightarrow C$ be a morphism such that $\varphi|_A$ is injective. Prove that φ is injective.
9. Determine the integral closure of $\mathbb{Z}[\sqrt{3}]$ in $\mathbb{Q}(\sqrt{3})$.
10. Determine the integral closure of $\mathbb{Z}[\sqrt{5}]$ in $\mathbb{Q}(\sqrt{5})$.

2.2 Assignment 2

Convention. By a ring, we mean a commutative ring with 1. Also, we exclude the zero ring.

1. Let $p \in \mathbb{Z}$ be a prime and write $\mathfrak{p} = \langle p \rangle$. Describe the elements of the local ring $\mathbb{Z}_{\mathfrak{p}}$. What is the field of fractions of $\mathbb{Z}_{\mathfrak{p}}$?

2. If N, P are submodules of an R -module M , prove that:

- (a) $S^{-1}(N + P) = S^{-1}N + S^{-1}P$;
- (b) $S^{-1}(N \cap P) = S^{-1}N \cap S^{-1}P$;
- (c) $S^{-1}(M/N) \cong S^{-1}M/S^{-1}N$ as $S^{-1}R$ -modules.

3. Let R be a ring and $S \subset R$ be a multiplicative subset. We have the canonical ring morphism $f : R \rightarrow S^{-1}R$ given by $f(r) = \frac{r}{1}$. For an ideal I of R , I^e denotes the ideal in $S^{-1}R$ generated by $f(I)$ (this is called *extension* of I). If $J \subseteq S^{-1}R$ is an ideal, then $f^{-1}(J)$ is an ideal of R which will be denoted as J^c (called the *contraction* of J).

- (a) Prove that any ideal of $S^{-1}R$ is an extended ideal.
- (b) Prove that an ideal I of R is contraction of an ideal of $S^{-1}R$ if and only if no element of S is a zero-divisor in R/I .
- (c) $\text{Spec}(S^{-1}R)$ is in bijective correspondence with $X = \{\mathfrak{p} \in \text{Spec}(R) \mid \mathfrak{p} \cap S = \emptyset\}$.
- (d) For ideals I, J of R , show that: $S^{-1}(I + J) = S^{-1}I + S^{-1}J$, $S^{-1}(I \cap J) = S^{-1}I \cap S^{-1}J$, $S^{-1}(IJ) = S^{-1}I S^{-1}J$, $S^{-1}(\sqrt{I}) = \sqrt{S^{-1}I}$.
- (e) If N is the nilradical of R then prove that the nilradical of $S^{-1}R$ is $S^{-1}N$.

4. Let $\mathfrak{p} \in \text{Spec}(R)$. Then $R_{\mathfrak{p}}$ is a local ring with unique maximal ideal $\mathfrak{p}R_{\mathfrak{p}}$ (we proved this). Therefore, $\frac{R_{\mathfrak{p}}}{\mathfrak{p}R_{\mathfrak{p}}}$ is a field, called the *residue field* at $\mathfrak{p}$. On the other hand, $R/\mathfrak{p}$ is an integral domain. Prove that its field of fractions is the same as $\frac{R_{\mathfrak{p}}}{\mathfrak{p}R_{\mathfrak{p}}}$.

5. Let M be a finitely generated R -module and S be a multiplicative subset of R . Prove that $S^{-1}(\text{Ann}(M)) = \text{Ann}(S^{-1}M)$.

6. Let M be a finitely generated R -module and S be a multiplicative subset of R . Prove that $S^{-1}M = 0$ if and only if there is $s \in S$ such that $sM = 0$.

7. Let J be an ideal of R . Show that $S := 1 + J := \{1 + a \mid a \in J\}$ is a multiplicative set. Prove that $S^{-1}J$ is contained in the Jacobson radical of $S^{-1}R$.

8. Let R be a ring and S, T be multiplicative subsets. Let U be the image of T in $S^{-1}R$. Define $ST := \{st \mid s \in S, t \in T\}$. Check that ST is also a multiplicative subset. Prove that the rings $(ST)^{-1}R$ and $U^{-1}(S^{-1}R)$ are isomorphic.

9. Let R be a ring such that $R_{\mathfrak{p}}$ has no nonzero nilpotent for every $\mathfrak{p} \in \text{Spec}(R)$. Prove that R has no nonzero nilpotent. If R is a ring such that $R_{\mathfrak{p}}$ is an integral domain for every $\mathfrak{p} \in \text{Spec}(R)$, then does it imply that R is an integral domain?

10. Let R be a (nonzero) ring and X be the set of all multiplicative subsets S of R such that $0 \notin S$. Show that X has a maximal element. Prove further that $T \in X$ is a maximal element of X if and only if $R \setminus T$ is a minimal prime ideal of R .

2.3 Assignment 3

Convention. By a ring, we mean a commutative ring with 1. Also, we exclude the zero ring.

1. Let R be an integral domain. Prove that the following are equivalent:
 - (a) R is integrally closed.
 - (b) $R_{\mathfrak{p}}$ is integrally closed for all $\mathfrak{p} \in \text{Spec}(R)$.
 - (c) $R_{\mathfrak{m}}$ is integrally closed for all $\mathfrak{m} \in \text{Max}(R)$.
2. Let $R \hookrightarrow S$ be an integral extension.
 - (a) Let $a \in R$ be such that a is a unit in S . Then show that a is a unit in R .
 - (b) Prove that $\text{Jac}(R) = \text{Jac}(S) \cap R$.
3. Let R be a ring and G be a finite subgroup of the group of automorphisms of R (automorphism means ring isomorphism from R to R , and they form a group with respect to composition). Let

$$R^G := \{r \in R \mid \sigma(r) = r \forall \sigma \in G\}.$$

- (a) Check that R^G is a subring of R and R is integral over R^G .
 - (b) Let $P \in \text{Spec}(R^G)$ and $X := \{Q \in \text{Spec}(R) \mid Q \cap R^G = P\}$. If $Q_1, Q_2 \in X$, prove that there is some $\tau \in G$ such that $Q_1 = \tau(Q_2)$. Conclude that X is finite.
4. Let $R \hookrightarrow S$ be an integral extension and I be an ideal of R . Prove that

$$\sqrt{IS} = \{s \in S \mid s^n + a_1 s^{n-1} + \cdots + a_n = 0 \text{ for some } n \in \mathbb{N} \text{ and } a_i \in I \forall i\}.$$

5. Let $R \hookrightarrow S$ be an integral extension and let $\mathfrak{n} \in \text{Max}(B)$. Let $\mathfrak{m} = \mathfrak{n} \cap A$ (note that $\mathfrak{m} \in \text{Max}(A)$). Is $B_{\mathfrak{n}}$ integral over $A_{\mathfrak{m}}$?
6. Let $R \hookrightarrow S$ be an extension of rings such that $B \setminus A$ is closed under multiplication. Prove that A is integrally closed in B .
7. Let R be a ring and $f \in R[X]$ be a nonconstant monic. Show that there is a ring S that contains R as a subring such that f is a product of monic linear factors in $S[X]$.
8. Let $R \hookrightarrow S$ be an extension of rings. Let f, g be two monic polynomials in $S[X]$. Let the coefficients of fg are all integral over R . Prove that the all coefficients of f and all the coefficients of g are integral over R .
9. Let $R \hookrightarrow S$ be an extension of rings. Prove that $f \in S[X]$ is integral over $R[X]$ if and only if all the coefficients of f are integral over R . Conclude that if C is the integral closure of R in S , then $C[X]$ is the integral closure of $R[X]$ in $S[X]$.

10. Let A be an integrally closed domain with field of fractions K . Let L be a finite Galois extension of K with Galois group G . Let B be the integral closure of A in L . Show that $\sigma(B) = B$ for all $\sigma \in G$ and that $A = B^G = \{b \in B \mid \sigma(b) = b \forall \sigma \in G\}$.

2.4 Assignment 4

Convention. By a ring, we mean a commutative ring with 1. Also, we exclude the zero ring.

1. Show that $2 \otimes \bar{1}$ is 0 in $\mathbb{Z} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{2\mathbb{Z}}$ but is not 0 in $2\mathbb{Z} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{2\mathbb{Z}}$.
2. Using the above exercise, or otherwise, prove that tensor product does not preserve exactness.
3. Let $a, b \in \mathbb{Z}$ be coprime. Prove that $\frac{\mathbb{Z}}{a\mathbb{Z}} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{b\mathbb{Z}} = 0$. More generally, if the gcd of a, b is d , prove that

$$\frac{\mathbb{Z}}{a\mathbb{Z}} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{b\mathbb{Z}} \cong \frac{\mathbb{Z}}{d\mathbb{Z}}.$$
4. Let R be a ring and I, J be ideals of R . Prove that $(R/I) \otimes_R (R/J) \cong R/(I+J)$.
5. Let M be an R -module and $I \subseteq R$ be an ideal. Prove that $(R/I) \otimes_R M \cong M/IM$.
6. Prove that $\mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} = 0$. Prove also that $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} = 0$.
7. Let M be an R -module and $I \subseteq R$ be an ideal. Prove that there is an R -linear map $g : I \otimes_R M \rightarrow IM$ and that g is surjective. Give an example to show that g is not injective in general.
8. Let $R \hookrightarrow S$ be rings. Prove that $S \otimes_R R[X]$ is isomorphic to $S[X]$ as R -modules.
9. Let R be an integral domain with quotient field K and let V be a K -vector space. Prove that $K \otimes_R V \cong V$ as R -modules.
10. Let M, N, P be R -modules. Prove that $M \otimes (N \otimes P) \cong M \otimes N \otimes P \cong (M \otimes N) \otimes P$.

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